The Bombay Salesian Society's
Don Bosco Institute of Technology, Mumbai
(An Autonomous Institute Affiliated to University of Mumbai)
Department of Electronics and Telecommunication Engineering

EXPERIMENT NO. 1

8-Bit Arithmetic operations using 8051

Aim: To write a program to perform addition, subtraction, multiplication and division of two 8bit numbers stored the result in the memory or register.

CO addressed: _____

Software used – Keil-vision 5

Learning Outcome:

1. Students will be able to understand various arithmetic instructions of 8051 microcontroller.
2. Students will be able to Perform arithmetic operations to observe status of parity, carry and overflow flags.

Theory:

Explain the instructions of 8051 microcontroller used for performing arithmetic operations

Instruction	Definition	General Syntax	Operation Performed	Flags Affected	Example
MOV	Transfers data from source to destination	MOV dest, src	dest ← src	None	MOV A,R0
ADD	Adds source to Accumulator	ADD A, src	A ← A + src	CY, AC, OV, P	ADD A,R1
SUBB	Subtracts source and borrow from A	SUBB A, src	A ← A - src - CY	CY, AC, OV, P	SUBB A,R1
MUL	Multiplies A and B	MUL AB	A × B → A (low), B (high)	OV (if >255), CY=0	MUL AB
DIV	Divides A by B	DIV AB	A/B → A=Quotient, B=Remainder	OV (if B=0), CY=0	DIV AB

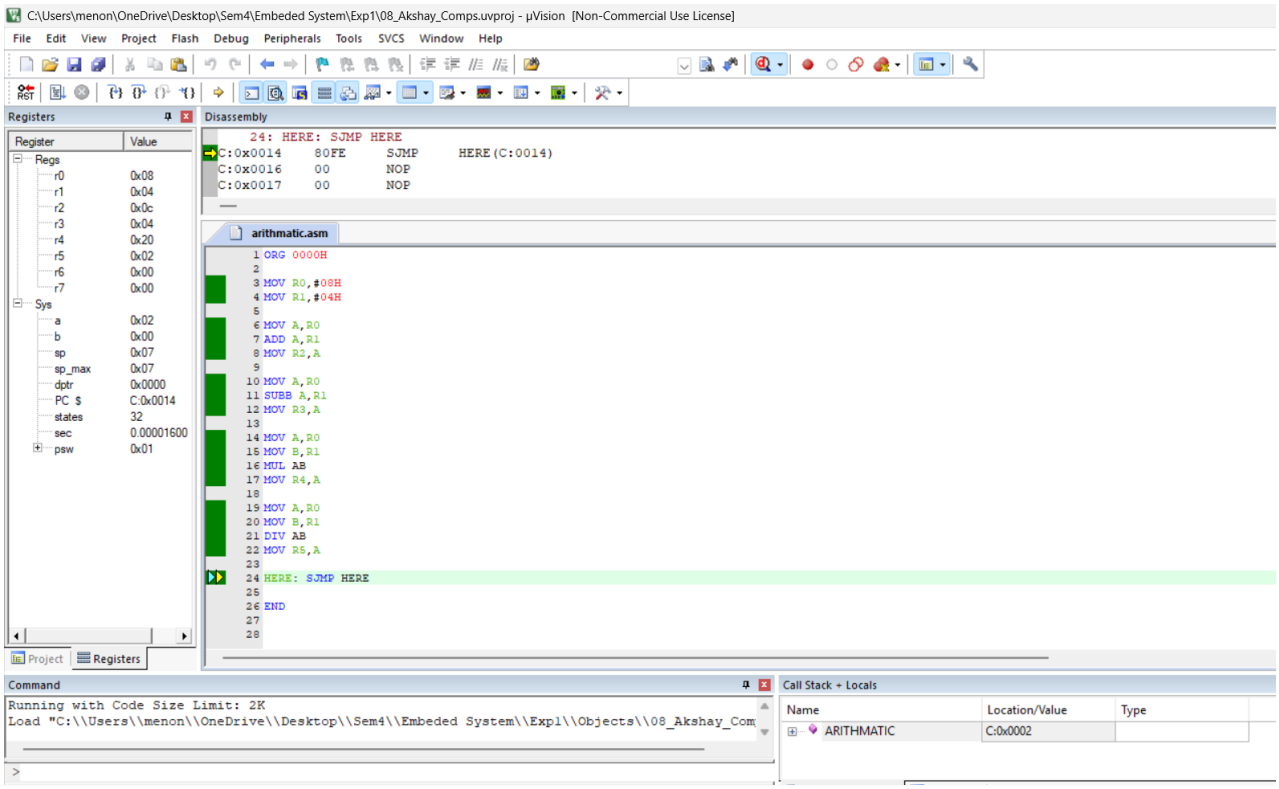
SJMP	Short jump to label	SJMP label	PC jumps to label	None	SJMP LOOP
END	Ends assembly program	END	Stops assembler reading program	None	END

Significance of HERE: SJMP HERE

1. It creates an infinite loop by continuously jumping to the same memory location.
2. It prevents the program from executing unwanted memory locations after completing execution.

Problem statement:

1. Write an Assembly language program to perform arithmetic operations on 8-bit numbers and store the results in the internal RAM



Conclusion:

In this experiment, 8-bit arithmetic operations such as addition, subtraction, multiplication, and division were successfully performed using the 8051 microcontroller. The operands were stored in registers/memory locations, and the results were computed using appropriate arithmetic instructions like ADD, SUBB, MUL AB, and DIV AB.

Through this experiment, we understood how the 8051 handles arithmetic operations internally, including the use of accumulator (A), B register, carry flag, and overflow conditions. It also helped in understanding register-level programming and the importance of flag bits in arithmetic computations.

Thus, the objective of performing 8-bit arithmetic operations and storing the results correctly in memory/registers was achieved successfully.